UNIVERSAL CANTOR SETS AND CANNON–THURSTON MAPS FOR FREE-BY-CYCLIC GROUPS

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In memory of Maxwell Plummer

Abstract. Given a hyperbolic free-by-cyclic group with irreducible monodromy and a component of its symmetrized BNS invariant, we construct a group action on a Cantor set that preserves a pair of laminations and that restricts to a uniform convergence group action by each associated fiber subgroup. As an application, there are only finitely many Cannon–Thurston maps for such a group that arise from fiber subgroups. We use the theory of dual laminations to $\mathbb{R}$-trees and Dowdall–Kapovich–Leininger's universal real tree in an approach analogous to the zippers machinery of Calegari–Loukidou. We additionally relate certain zippers we construct to some of their zippers.

1. Introduction

Cannon and Thurston famously proved that the inclusion of a fiber surface S into a hyperbolic 3-manifold M gives rise to a continuous $\pi_1(M)$-equivariant surjection $S^1 \to S^2$ [CT07]. Given a normal hyperbolic subgroup H of a hyperbolic group G, the action of H on its Gromov boundary $\partial_\infty H$ extends to an action of G on $\partial_\infty H$. Mj (formerly Mitra) proved that in this more general setting the inclusion of H into G defines a continuous G-equivariant *Cannon–Thurston map* $i\colon \partial_\infty H \to \partial_\infty G$ [Mit98].

A *free-by-cyclic group* G is the mapping torus group of some $\varphi \in \mathrm{Out}(F_N)$, that is, $G \cong F_N \rtimes_\varphi \mathbb{Z}$. When G is hyperbolic, Mj's theorem guarantees a Cannon–Thurston map $\partial_\infty F_N \to \partial_\infty G$. If the first Betti number of G is at least two, there are infinitely many ways of writing G as such a semidirect product; these ways are governed by the *symmetrized BNS invariant* $\Sigma_s(G) = \Sigma(G) \cap -\Sigma(G)$ [BNS87]. Thus, there are infinitely many Cannon–Thurston maps for a fixed free-by-cyclic group G; a study of the relationship between these Cannon–Thurston maps was proposed by Kapovich–Lustig [KL15]. Our main theorem is that when G has *irreducible monodromy*, up to G-equivariant homeomorphism, the Cannon–Thurston maps only depend on a connected component of $\Sigma_s(G)$.

Theorem A. *Let G be the mapping torus group of a fully irreducible atoroidal outer automorphism $\varphi \in \mathrm{Out}(F_N)$. Let Σ_0 be a component of the symmetrized BNS invariant $\Sigma_s(G)$. There is a G-action on a Cantor set $\mathfrak{C}$ preserving a pair of algebraic laminations and a G-equivariant map $\mathcal{R}\colon \mathfrak{C} \to \partial_\infty G$ such that, for each rational $u \in \Sigma_0$, there is a G-equivariant homeomorphism $h_u\colon \partial_\infty \ker u \to \mathfrak{C}$ such that the Cannon–Thurston map $i_u\colon \partial_\infty \ker u \to \partial_\infty G$ factors as $i_u = \mathcal{R} \circ h_u$.*

The G-equivariant identification of $\ker u$ with $\mathfrak{C}$ ensures, via a theorem of Bowditch [Bow98], that the action of $\ker u$ on $\mathfrak{C}$ is a *uniform convergence group action*. We refer to

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the action of G on $\mathfrak{C}$ as a *universal Cantor set*, in analogy with the *universal circles* that arise in 3-manifold theory. It is a theorem of Kielak that $\Sigma_s(G)$ has finitely many components [Kie20], we thus have have the following corollary:

Corollary 1.1. *For G as in Theorem A, up to G-equivariant homeomorphism, there are finitely many equivalence classes of Cannon–Thurston maps $i_u\colon \ker u \to \partial_\infty G$, where $u\colon G \to \mathbb{Z}$ has finitely generated kernel.*

We briefly describe our construction of $\mathfrak{C}$, which is analogous to Calegari–Loukidou's construction of a universal circle from a *zipper* [CL26b]. Given a monodromy φ, we look at the image $Z^\pm$ of the attracting and repelling $\mathbb{R}$-trees $T_\pm$ in $\partial_\infty G$. The Cantor set $\mathfrak{C}$ consists of (a large subset of) the end spaces of T_+ and T_-, modulo when an end of T_+ and an end of T_- determine the same limit point in $\partial_\infty G$. The identification of $\mathfrak{C}$ with $\partial_\infty G$ is produced by an analysis of certain equivariant maps $\mathcal{Q}_\pm\colon \partial_\infty F_N \to \widehat{T}_\pm$, where $\widehat{T}_\pm = \overline{T}_\pm \cup \partial T_\pm$, where $\overline{T}_\pm$ is the metric completion of T, see [LL03]. Recognizing that the Cantor set $\mathfrak{C}$ is independent of the monodromy chosen can be seen via Dowdall–Kapovich–Leininger's *universal real trees* $T^{\mathrm{univ}}_\pm$—defined in the forthcoming [DKL26]—which each depend only on a component of the BNS invariant. In fact, $Z^\pm$ can be seen to be images of $T^{\mathrm{univ}}_\pm$, and so $\mathfrak{C}$ depends only on the component of $\Sigma(G) \cap -\Sigma(G)$.

1.1. **Universal circles and the universal Cantor set.** The notion of a *universal circle* arises in various forms in the study of 3-manifolds. Roughly, in this setting, a universal circle is an action of $\pi_1(M)$ on S^1 that naturally arises from a geometric or dynamical structure on M. They first arose (non-uniquely) from taut foliations, discovered by Thurston and written down and extended in a paper of Calegari–Dunfield [CD03]. For taut foliations, universal circles consist mainly of a collection of monotone maps from circles associated to leaves of the foliation to the universal circle in a way that intertwines the actions.

Given a closed hyperbolic 3-manifold M, we may also construct universal circles for M from pseudo-Anosov flows [CD03; Fen12], and more generally quasi-geodesic flows [Cal06; Fra15]. These universal circle actions preserve a pair of circle laminations: in the suspension flow setting these are the invariant laminations of the pseudo-Anosov homeomorphism, see [CB88]. While universal circles for flows lack the axiomatic description for foliations, work of Landry–Minsky–Taylor [LMT26] shows that for any taut foliation $\mathcal{F}$ transverse to a pseudo-Anosov ψ flow one can treat the universal circle of the orbit space S^1_ψ as also a universal circle for $\mathcal{F}$. Universal circles simplify the dynamics of the flow from a continuous action (that of $\mathbb{R} \curvearrowright M$ given by the flow) to a discrete one (the action of $\pi_1 M \curvearrowright S^1_u$).

We view our universal Cantor set action as more analogous to the universal circles arising from a pseudo-Anosov flow: Dowdall–Kapovich–Leininger's universal real tree is roughly the orbit space of a semiflow associated to a component of $\Sigma(G)$. Fried showed that a fibered face of the Thurston norm ball [Thu86]—equivalently, a component of the BNS invariant—is associated to a single pseudo-Anosov suspension flow [Fri82], see also [FLP12, Exposé 14]. The association of $\mathfrak{C}$ to a component of $\Sigma(G) \cap -\Sigma(G)$ rather than to $\Sigma(G)$ is indicative of the fact that we need a pair of two 'opposite' semiflows to construct $\mathfrak{C}$. While we will not dwell on this point, there is also a taut foliations perspective for $\mathfrak{C}$: associated to any $u \in \Sigma(G) \cap -\Sigma(G)$, there is a 'foliation' of a 2-complex X by fibers [DKL15; DKL17b]. When u is rational, the Gromov boundary of the leaves of the lifted foliation to $\tilde{X}$ are all identified to $\partial_\infty F_N$, and hence with $\mathfrak{C}$. We expect that the same can be said for foliations associated to irrational u, but this does not follow from the present work.

1.2. **Zippers.** First introduced in [CL26b], a *zipper* for a hyperbolic group G is a pair of disjoint subsets $Z^{\pm} \subset \partial_\infty G$ such that each of $Z^{\pm}$ are nonempty, path-connected, and G-invariant. In [CL26b], Calegari–Loukidou mostly consider the case where $G = \pi_1(M)$ for M a closed, hyperbolic manifold or that $\partial_\infty G \cong S^2$. On S^2, using the Jordan curve theorem, they show that $Z^{\pm}$ are *real trees* and that one may construct a universal circle S^1_{CL} by completing the natural circle orders of the end spaces of $Z^{\pm}$ inherited from S^2. The prototypical example of a zipper is as the forward or backward endpoints of orbits of a quasi-geodesic or pseudo-Anosov flow with no perfect fits.

The case where $\partial_\infty G \ncong S^2$ is relatively unexplored. Calegari–Loukidou show that given a *uniform quasimorphism* $u\colon G \to \mathbb{R}$, there is a zipper $Z^{\pm}(u) \subset \partial_\infty G$. From such a quasimorphism, they associate certain quasi-geodesics $S\colon \mathbb{Z} \to G$ called *staircases*, which each determine a point $p_S \in Z^+(u) \subset \partial_\infty G$. The set $Z^+_{CL}(u)$ is the set of all endpoints of all such staircases and $Z^-_{CL}(u)$ is the set of endpoints of staircases associated to $-u$. Our zippers $Z^{\pm}$ are constructed as the endpoints of a pair of semiflows associated to a component of $\Sigma_s(G)$, or alternatively as the images of two the universal real trees in $\partial_\infty G$. Our second theorem relates the two constructions of zippers:

Theorem B. *Let G be a hyperbolic free-by-cyclic group with irreducible monodromy. Let Σ_0 be a component of $\Sigma_s(G)$ and $Z^{\pm}$ the zippers arising from the universal real trees of Σ_0 and $-\Sigma_0$.*

For each surjective $u\colon G \to \mathbb{Z}$ with $u \in \Sigma_0$, we have that $Z^{\pm}(\Sigma_0)$ is a minimal subzipper of $Z^{\pm}_{CL}(u)$. Further, if $b_1(G) \geq 2$, the containment is proper and $Z^+_{CL}(u)$ is not a minimal zipper.

See Propositions 6.5 and 6.6 for full statements, which necessitate some care with certain constants involved. To our knowledge, this is the first consideration of minimality of zippers for a hyperbolic group G with $\partial_\infty G \ncong S^2$. In particular, the fact that the same notion of minimality that applies in S^2 is reasonable in another setting appears to be new. We additionally remark that the proof of Theorem B can be stated almost without modification in the setting where $Z^{\pm}$ are the endpoints of flow lines of a pseudo-Anosov suspension flow.

Outline. In Section 2, we give a brief review of $\mathrm{Out}(F_N)$ and $\mathbb{R}$-trees, and discuss their topological generalizations *real trees*. In Section 3, we introduce the necessary aspects of free-by-cyclic groups and define Dowdall–Kapovich–Leininger's universal real tree. We return to $\mathrm{Out}(F_N)$ in Section 4, where we discuss several related algebraic laminations and the equivariant map $\mathcal{Q}$. We construct the universal Cantor set $\mathfrak{C}$ in Section 5 and prove Theorem A. Finally, in Section 6 we discuss the zippers perspective and prove Theorem B.

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2. $\mathrm{Out}(F_N)$ and real trees

We follow Bestvina–Handel [BH92] in this discussion. An outer automorphism $\varphi \in \mathrm{Out}(F_N)$ is *fully irreducible* if no power of φ fixes the conjugacy class of a proper free factor of F_N. Similarly, φ is *atoroidal* if no conjugacy class of a nontrivial element of F_N is fixed by a power of φ.

By a graph Γ, we will mean a 1-dimensional cell complex without valence 1 vertices. A *train track map* $f\colon \Gamma \to \Gamma$ is a function that (1) sends vertices to vertices (2) maps edges over

edge paths, and (3) is such that for all n, f^n is locally injective on the interior of each edge. A train track map f is *irreducible* if for each pair of edges e, e' of Γ, there is a power n such that $f^n(e)$ maps over e'; f is *expanding* if for any edge e, the simplicial edge length of $f^n(e)$ goes to infinity as n increases. A train track map *topologically represents* $\varphi \in \mathrm{Out}(F_N)$ if $\pi_1(\Gamma)$ is identified with F_N so that f_* induces φ. If $\varphi \in \mathrm{Out}(F_N)$ is fully irreducible, it is topologically represented by an expanding irreducible train track map [BH92].

2.1. **Outer Space and Attracting Trees.** A *simplicial tree* is a simply-connected 1-complex and a *metric simiplicial tree* is a geodesic metric on a simplicial tree. An $\mathbb{R}$-*tree* is a connected geodesic metric space that has unique arcs (embedded intervals) between any pair of points, see [Bes02] for a nice overview on the theory of $\mathbb{R}$-trees.

The *outer space* CV_N is the space of free F_N actions on metric simplicial trees [CV86] and its compactification $\overline{CV}_N$ is the space of *very small* F_N actions on $\mathbb{R}$-trees [CM87; BF94]. In both cases trees T_1, T_2 are equivalent if there is an F_N-equivariant homothety $h\colon T_1 \to T_2$, in particular, the metrics are considered only up to positive scaling. The group $\mathrm{Out}(F_N)$ has a right action on $\overline{CV}_N$ by precomposition of the action. A fully irreducible $\varphi \in \mathrm{Out}(F_N)$ acts on $\overline{CV}_N$ with north-south dynamics [LL03], fixing two $\mathbb{R}$-trees $T_\pm = T_\pm(\varphi)$ in ∂CV_N [BFH97; BFH00; GJLL98].

The *attracting tree* $T_+(\varphi)$ can be explicitly constructed from an expanding irreducible train track representative $f\colon \Gamma \to \Gamma$ [BFH97]. The graph Γ admits a metric arising from the Perron–Frobenius eigenvector of its transition matrix under which f can be taken to uniformly expand edge lengths by the Perron–Frobenius eigenvalue $\lambda > 1$; λ is also called the *stretch factor* of f or of φ. Let $\tilde{\Gamma}$ be the metric universal cover, which is a metric simplicial tree, and let $\tilde{f}\colon \tilde{\Gamma} \to \tilde{\Gamma}$ be a lift of f. Define a pseudo-metric on $\tilde{\Gamma}$ by

$$d_\infty(x, y) = \lim_{n\to\infty} \frac{d(\tilde{f}^n(x), \tilde{f}^n(y))}{\lambda^n}.$$

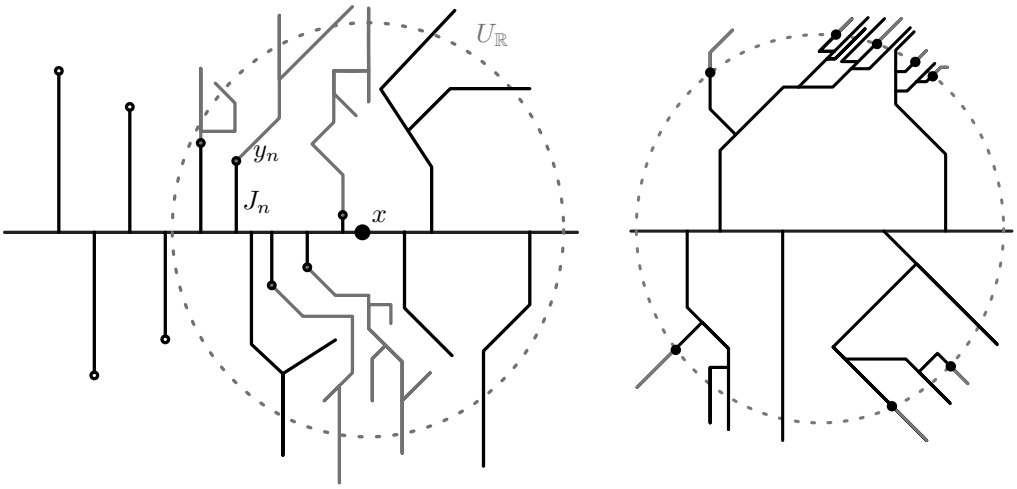
Declaring $x \sim y$ if $d_\infty(x, y) = 0$, d_∞ descends to a metric on the attracting tree $T_+ = T_+(\varphi) = \tilde{\Gamma}/\sim$. The *repelling tree* $T_- = T_-(\varphi)$ is equal to $T_+(\varphi^{-1})$. The action of F_N on T_+ and T_- is *minimal* in the sense that there is no proper invariant subtree, and the *branch points* of $T_\pm$ are dense.

In the construction of the attracting tree, the map $\tilde{f}$ induces a map $H\colon T_+ \to T_+$; a calculation using the definition above shows that H acts as a λ-homothety on T_+, see e.g. [GJLL98]. In this way, we can extend the isometric action of F_N on T_+ to an action of $G = F_N \rtimes_\Phi \mathbb{Z} = \langle w, r \mid w \in F_N, r^{-1}wr = \Phi(w)\rangle$ by homotheties, where $\Phi \in \mathrm{Aut}(F_N)$ is a representative of φ, with r^{-1} acting as H.

We remark that Bestvina–Feighn–Handel also showed that a fully irreducible $\varphi \in \mathrm{Out}(F_N)$ preserves a *stable lamination* $L_{\mathrm{BFH}}(\varphi)$ and *unstable lamination* $L_{\mathrm{BFH}}(\varphi^{-1})$ [BFH97]. The lamination $L_{\mathrm{BFH}}(\varphi)$ is obtained from a train track representative $f\colon \Gamma \to \Gamma$ by iteration on the interior of edges. We will discuss laminations more in Section 4.

2.2. **Real Trees.** We will refer to an arc in a topological space either as an embedding of an interval or as the image of such an embedding. A topological space $(X, \mathcal{T})$ is *unique arc connected* if, for each pair of points $x, y \in X$, there is a unique arc, which we denote $[x, y]$, connecting them. In this section, we will be careful to specify the topology on a set; we will mostly revert to not doing so in future sections.

Definition 2.1. A *real tree* is a topological space $(X, \mathcal{T})$ that is Hausdorff, locally path-connected, and unique-arc connected.



(A) An open set $U_{\text{int}} \in \mathcal{T}_{\text{int}}$ may exclude points at arbitrarily small distance from a point x, so x has no $\mathcal{T}_{\mathbb{R}}$-neighborhood contained in U_{int}. The grey lines represent arcs in $U_{\mathbb{R}} \in \mathcal{T}_{\mathbb{R}}$ which are not in U_{int}.

(B) A metric ball $U_{\mathbb{R}} \in \mathcal{T}_{\mathbb{R}}$ may be complicated enough to not contain any finite intersection of directions $U \in \mathcal{T}_{\text{obs}}$.

FIGURE 1. Schematic depicting the distinct topologies $\mathcal{T}_{\text{obs}}, \mathcal{T}_{\mathbb{R}}$, and $\mathcal{T}_{\text{int}}$.

Real trees are sometimes referred to in the literature as *topological* $\mathbb{R}$*-trees*, indeed, $\mathbb{R}$-trees are the prototypical examples of a real tree. The name used here is that used in [DKL26], following Bowditch's systematic treatment of tree-like spaces [Bow99].

If $(X, \mathcal{T})$ is a real tree and $x \in X$, a *direction* at x is a connected component of $X \setminus \{x\}$. A point $x \in X$ is a *branch point* if there are at least three directions at x. Let $\mathcal{D}(X, \mathcal{T})$ be the collection of all directions. The following two elementary lemmas will establish that a direction is also a path-component of $X \setminus \{x\}$.

Lemma 2.2 ([Bow99], Lemma 1.5)**.** *If $D \subset X$ is a direction at a point $x \in X$, the $D \in \mathcal{T}$.*

Proof. Since X is Hausdorff, $X \setminus \{x\}$ is open. Since X is locally connected, a component of $X \setminus \{x\}$ is open. □

Lemma 2.3 ([Bow99], Lemma 1.4)**.** *Let $(X, \mathcal{T})$ be a real tree. If $A \subseteq X$ is connected, A is path-connected.*

Proof. Suppose $A \subseteq X$ is not path-connected. There are points $x, y \in X$ such that $[x, y] \nsubseteq A$. Let $z \in [x, y] \setminus A$ and D be the direction at z containing x. Then $x \in A_1 = D \cap A$ and $y \in A_2 = A \setminus D$, and A_1 and A_2 are both open, so A is not connected. □

Definition 2.4. Let $(X, \mathcal{T})$ be a real tree. The *intervals topology* $\mathcal{T}_{\text{int}}$ on X is the collection

$$\mathcal{T}_{\text{int}} = \{U \subseteq X \mid U \cap J \text{ is open for all arcs } J \subseteq X\}.$$

The *observers' topology* $\mathcal{T}_{\text{obs}}$ on X is the topology with subbasis $\mathcal{D}(X, \mathcal{T})$, that is, finite intersections of directions form a basis of the observers' topology.

Additionally, if $\mathcal{T}_{\mathbb{R}}$ is the topology on a tree arising from an $\mathbb{R}$-tree metric on X, we refer to $\mathcal{T}_{\mathbb{R}}$ as the *metric topology*.

Example 2.5. We will show that if $(X, \mathcal{T}_{\mathbb{R}})$ is an $\mathbb{R}$-tree with dense branch points, the topologies $\mathcal{T}_{\text{int}}, \mathcal{T}_{\mathbb{R}}, \mathcal{T}_{\text{obs}}$ are typically distinct, see Figure 1.

Let $[x, y]$ be an arc with a sequence of pairwise distinct branch points x_n converging to x. For each n, let J_n be an arc starting at x_n with length $1/n$ with $J_n \cap [x, y] = \{x_n\}$. Let $U_{\mathbb{R}} \in \mathcal{T}_{\mathbb{R}}$ be a connected neighborhood of x. For each n, let y_n be on the interior of the arc J_n, let U_{int} be the connected component of $U_{\mathbb{R}} \setminus \{y_n \mid n \in \mathbb{N}\}$ containing x. The set U_{int} is in $\mathcal{T}_{\text{int}} \setminus \mathcal{T}_{\mathbb{R}}$: each open metric ball around x contains infinitely many y_n, but no y_n is in U_{int}, thus $U_{\text{int}} \notin \mathcal{T}_{\mathbb{R}}$.

For the observers' topology, we note that for any metric ball $U_{\mathbb{R}} \in \mathcal{T}_{\mathbb{R}}$, there are typically countably many points of the $\mathcal{T}_{\mathbb{R}}$-frontier $\text{fr}_{\mathbb{R}}(U_{\mathbb{R}})$. Thus no finite intersection of directions will be contained in $U_{\mathbb{R}}$, and $U_{\mathbb{R}} \notin \mathcal{T}_{\text{obs}}$.

Before we go on proving a number of simple but technical lemmas, we briefly describe the significance of these topologies to our work here. The observers' topology was introduced in [CHL07], and will be the key tool in establishing continuity of the map we construct from the boundary of a fiber group to the Cantor set $\mathfrak{C}$ we construct later.

The intervals topology was introduced in [DKL26] and helps to relate the attracting $\mathbb{R}$-trees for monodromies within a component of the BNS invariant. This will be the tool we use to relate the Cantor set actions across a component of the symmetrized BNS invariant.

The purpose of the rest of this section is to bridge the gap between the typically distinct intervals and observers' topologies. Given two real tree topologies $\mathcal{T}, \mathcal{T}'$ on X, we say $\mathcal{T}$ and $\mathcal{T}'$ *have the same arcs* if an embedded arc of $\mathcal{T}$ is an embedded arc of $\mathcal{T}'$ and vice versa. We note that it is established in [DKL26] that $(X, \mathcal{T}_{\text{int}})$ is a real tree topology with the same arcs as $(X, \mathcal{T})$. For the observers' topology, [CHL07] proved it in a special case; we will prove it here as well.

Proposition 2.6. *Let $(X, \mathcal{T})$ be a real tree and $\mathcal{T}_{obs}$ be the observers' topology on X. The space $(X, \mathcal{T}_{obs})$ is a real tree with the same arcs as $(X, \mathcal{T})$.*

We remark that, given a real tree $(X, \mathcal{T})$, the observers' topology is the coarsest real tree topology on X with the same arcs as $(X, \mathcal{T})$. Similarly, the intervals topology on X is the finest real tree topology with the same arcs as $(X, \mathcal{T})$.

Proof. Given $x \neq y \in X$, there is a point z in the open $\mathcal{T}$-arc (x, y), and x, y are in different directions at z. Thus, $(X, \mathcal{T}_{\text{obs}})$ is Hausdorff.

By definition, $\mathcal{T}_{\text{obs}}$ admits a subbasis consisting of directions. Say $U \in \mathcal{T}_{\text{obs}}$ is a finite intersection of directions. By Lemma 2.2, $U \in \mathcal{T}$. Since $(X, \mathcal{T})$ is unique-arc connected, U is $\mathcal{T}$-connected, thus $\mathcal{T}$-path-connected, so for any $x, y \in U$, there is a $\mathcal{T}$-arc $[x, y]$ between them. The identity map $\text{id}\colon (X, \mathcal{T}) \to (X, \mathcal{T}_{\text{obs}})$ is continuous, so $[x, y]$ is a $\mathcal{T}_{\text{obs}}$-arc as well. We have established local path-connectedness.

Let $x, y \in X$ and $[x, y]$ the $\mathcal{T}$-arc between them. We have established that $[x, y]$ is a $\mathcal{T}_{\text{obs}}$-arc as well. Let $\gamma\colon I \to X$ be a $\mathcal{T}_{\text{obs}}$-arc with $\gamma(0) = x$. Let $J = \gamma(I) \cap [x, y]$, suppose there is a $z \in J \setminus [x, y]$ and let D be the direction at z containing x. Since $z \notin [x, y]$, $[x, y] \subseteq D$ and the endpoint of γ is not in D, so γ is not a $\mathcal{T}_{\text{obs}}$-arc between x and y. A similar argument shows that if there is a point $z \in [x, y] \setminus J$, then $[x, y]$ is not a $\mathcal{T}_{\text{obs}}$-arc between x and $\gamma(1)$, so $(X, \mathcal{T}_{\text{obs}})$ is unique arc connected with the same arcs as $(X, \mathcal{T})$. □

The *convex hull* of a subset $A \subset X$ is the union

$$\text{hull}(A) = \bigcup_{x,y \in A} [x, y].$$

A *finite subtree* of a real tree is the convex hull of a finite set.

Definition 2.7. Let $(X, \mathcal{T})$ be a separable real tree; it is a countable union of finite subtrees T_i. We will refer to its *end space* $E(X)$ as the space

$$\varprojlim \pi_0(X \setminus T_i).$$

Explicitly, let $\mathcal{I}$ be a countable set indexing i, topologize $\pi_0(X \setminus T_i)$ with the discrete topology, and consider the space $\prod_{i\in\mathcal{I}} \pi_0(X \setminus T_i)$. A point $(D_i)_{i\in\mathcal{I}}$ of this product is a member of $E(X)$ if and only if for any two $i, j \in \mathcal{I}$, $D_i \cap D_j \neq \varnothing$. We give $E(X)$ the subspace topology. One can check that the set of directions on X determines a basis of $E(X)$. In practice, we will use a *finite subtree exhaustion* when discussing the end space of a particular real tree.

We note that we cannot replace finite subtrees in the above with compact sets, as we do not wish to restrict our attention to locally compact real trees.

Lemma 2.8. *Let X be a set and $\mathcal{T}_1, \mathcal{T}_2$ topologies on X such that each of $(X, \mathcal{T}_i)$ is a real tree. If $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ have the same arcs, the identity map on X induces a homeomorphism of the end spaces of $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$.*

Proof. Since $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ have the same arcs, they have the same path-connected subsets, so they have the same connected subsets by Lemma 2.3. A finite subtree is defined solely in terms of arcs, so $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ have the same finite subtrees. If $T \subset X$ is a finite subtree, a connected component of $X \setminus T$ is a direction at a point of T, so $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ share the same connected components of complements of finite subtrees. Thus, an end of $(X, \mathcal{T}_1)$ uniquely determines an end of $(X, \mathcal{T}_2)$ in a continuous manner, and vice versa. □

Given an $\mathbb{R}$-tree $T = (X, \mathcal{T}_\mathbb{R})$, let $\overline{T} = (\overline{X}, \overline{\mathcal{T}}_\mathbb{R})$ denote its metric completion and $\widehat{T} = \overline{T} \cup \partial_\infty T = (\widehat{X}, \widehat{\mathcal{T}}_\mathbb{R})$ be $\overline{T}$ together with its Gromov boundary: the space of asymptote classes of geodesic rays. The space $\overline{T}$ is not necessarily proper, so $\widehat{T}$ may not be compact. Instead, we will consider the observers' topology on $\widehat{T}$, $\widehat{T}_{\text{obs}} = (\widehat{X}, \widehat{\mathcal{T}}_{\text{obs}})$.

Lemma 2.9. *If $(X, \mathcal{T}_\mathbb{R})$ is separable, the space $\widehat{T}_{obs} = (\widehat{X}, \widehat{\mathcal{T}}_{obs})$ is a* dendrite*—a compact separable real tree.*

Proof. The compactness conclusion is given in [CHL07, Proposition 1.13]; we will show that $\widehat{T}_{\text{obs}}$ is unique-arc connected. Let $x, y \in \widehat{X}$ and $(T_n)_{n\in\mathbb{N}}$ be a finite subtree exhaustion of X. We define a sequence $x_n \in X$ as follows:

(1) If $x \in T_n$, $x_n = x$.
(2) If $x \in \widehat{X} \setminus T_n$, let U_n be the component of $\widehat{X} \setminus T_n$ containing x, and define x_n to be the unique point of $T_n \cap \overline{U}_n$, with the closure taken with respect to $\mathcal{T}_\mathbb{R}$. Equivalently, if $\gamma\colon [0, 1] \to X$ is an arc with $\gamma(0) \in T_n$ and $\gamma(1) \in U_n$, x_n is the last point of intersection of γ with T_n.

Likewise, we define y_n. In this way we obtain an increasing sequence of unique arcs $[x_n, y_n]$ that extends to an arc $[x, y]$. For uniqueness, if $\gamma : [0, 1] \to \widehat{X}$ is an arc with $\gamma(0) = x, \gamma(1) = y$, then $\gamma((0, 1)) \subseteq X$ and the image of γ agrees with $[x, y]$ on each finite subtree of X; in particular the interior arcs agree and extend to $\widehat{X}$ in the same way. □

Lemma 2.10. *Let $(X, \mathcal{T})$ be a real tree and $f\colon X \to X$ a $\mathcal{T}$-homeomorphism. The map f is also a $\mathcal{T}_{obs}$-homeomorphism.*

Proof. For any $x \in X$, f is a homeomorphism $(X \setminus \{x\}, \mathcal{T}) \to (X \setminus \{f(x)\}, \mathcal{T})$, so f sends directions to directions. The same is true for f^{-1}, so f is a homeomorphism of $\mathcal{T}_{\text{obs}}$. □

Proposition 2.11. *Let G be a group acting on a separable $\mathbb{R}$-tree $T = (X, \mathcal{T})$ by homeomorphisms. There is a G-equivariant homeomorphism $(\widehat{T} \setminus T)_{obs} \to E(T)$.*

Proof. Let $\mathfrak{R}$ be the set of non-extendible properly embedded rays $\gamma\colon [0, \infty) \to T$, where γ, γ' are equivalent if their images agree outside of a finite subtree. The group G acts on the set $\mathfrak{R}$, and there are G-equivariant identifications of $\mathfrak{R}$ with $\widehat{T} \setminus T$ and $E(T)$. We will show that the induced topologies $\mathcal{T}_{\text{obs}}, \mathcal{T}_{\text{ends}}$ on $\mathfrak{R}$ are the same.

Given a direction D of $(X, \mathcal{T})$ at a point $p \in X$, let $U_D \in \mathcal{T}_{\text{obs}}$ be the collection of rays in $\mathfrak{R}$ with infinite intersection with D. The set $\{p\}$ is a finite subtree, and D is a complementary component of $\{p\}$, so $U \in \mathcal{T}_{\text{ends}}$. On the other hand, a complimentary component of a finite subtree is a direction at a point of the finite subtree, so $\mathcal{T}_{\text{ends}} \subseteq \mathcal{T}_{\text{obs}}$. □

We note that in general, the topology on $\widehat{T} \setminus T$ coming from the $\mathbb{R}$-tree topology is distinct from the observers' topology. For instance, $\overline{T} \setminus T \subseteq \widehat{T} \setminus T$ is $\mathcal{T}_{\mathbb{R}}$-open, but in many cases is not $\mathcal{T}_{\text{obs}}$-open.

3. Free-by-cyclic groups and the universal real tree

3.1. **Free-by-cyclic groups and their BNS invariants.** Let $G \cong F_N \rtimes_\varphi \mathbb{Z}$ be a free-by-cyclic group. If the first Betti number of G is at least 2, G admits infinitely many surjective homomorphisms $G \to \mathbb{Z}$, which we view as the primitive integral points of $\mathrm{Hom}(G; \mathbb{R})$. Let $S(G) = (\mathrm{Hom}(G; \mathbb{R}) \setminus \{0\})/\mathbb{R}_+$; Bieri–Neumann–Strebel defined the *BNS invariant* $\Sigma(G)$, which is a certain open subset of $S(G)$ [BNS87]. If $u\colon G \to \mathbb{Z}$ is surjective, u has finitely generated kernel if and only if $[u] \in \Sigma(G) \cap -\Sigma(G)$; each such finitely generated kernel is free [GMSW01]. For $u\colon G \to \mathbb{Z}$ with $[u] \in \Sigma(G) \setminus -\Sigma(G)$, u is dual to a splitting of G as a strictly ascending HNN-extension: G is the mapping torus group of an injective nonsurjective endomorphism of a finitely generated free group. Kielak proved that the BNS invariant of a free-by-cyclic group is polyhedral [Kie20], so it is determined by its rational points. Going forward, we will abuse notation slightly and write $u \in \Sigma(G)$ whenever $u \in \mathrm{Hom}(G; \mathbb{R}) \setminus \{0\}$ projects into $\Sigma(G)$.

In the case where $b_1(G) \geq 2$, G splits as a free-by-cyclic group in infinitely many ways with infinitely many monodromies. Some properties of the monodromies only depend on the group: Brinkmann proved that G is hyperbolic if only if the monodromy is atoroidal [Bri00], see also [BF92]. Mutanguha proved that (full) irreducibility is an invariant of the group [Mut21], building on work of Dowdall–Kapovich–Leininger, who showed it was an invariant of a component of the BNS invariant [DKL17a]. We will be concerned with hyperbolic free-by-cyclic groups with irreducible monodromy, that is, some (any) monodromy $\varphi \in \mathrm{Out}(F_N)$ with $G \cong F_N \rtimes_\varphi \mathbb{Z}$ is both atoroidal and fully irreducible.

3.2. **The mapping torus and universal real tree.** We follow Dowdall–Kapovich–Leininger in this discussion, especially [DKL17b], see also [DKL15].

Fix a hyperbolic free-by-cyclic group $G \cong F_N \rtimes_\varphi \mathbb{Z}$ with irreducible monodromy φ. Explicitly, choose some $\Phi \in \mathrm{Aut}(F_N)$ representing φ, we can write $G \cong \langle a_1, \dots, a_N, r \mid r^{-1} a_n r = \Phi(a_n) \rangle$.

Let $f\colon \Gamma \to \Gamma$ be an expanding irreducible train track map representing φ and $X_f = \Gamma \times [0,1]/(x,1) \sim (f(x), 0)$ its mapping torus, which is a $K(G, 1)$. The space X_f comes equipped with a suspension semiflow $\psi\colon X \times [0, \infty) \to X$, locally defined by $\psi_s, (x, t) = (x, t+s)$ when

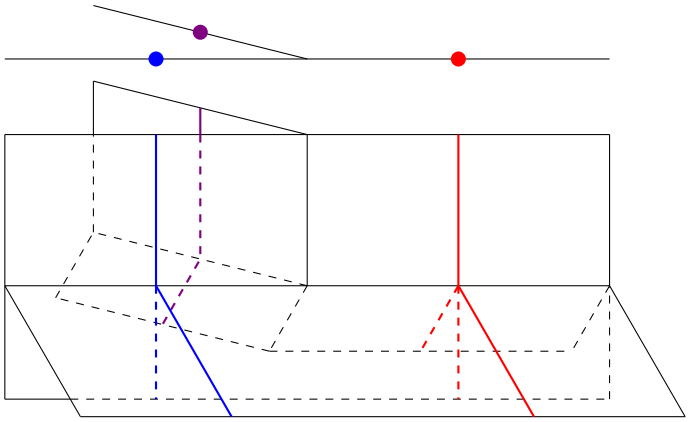


FIGURE 2. At the top is a piece of the universal real tree. Below is a portion of $\tilde{X}$, the universal cover of the mapping torus of f. The sets in color are the preimages of the colored points in the tree under the quotient map $\tilde{X} \to \tilde{X}/\sim$.

$\leq s, t \leq s+t \leq 1$; we will typically write $\psi_t(x) = \psi(x,t)$. The monodromy φ is dual to a map $u\colon G \to \mathbb{Z}$ with kernel F_N and so defines a component Σ_0 of $\Sigma(G)$. For any surjective $u\colon G \to \mathbb{Z}$ with $u \in \Sigma_0$, there is a cross section $\Theta_u \subset X$ of ψ with well-defined first return map $f_u\colon \Theta_u \to \Theta_u$ that is a train track map. The map f_u represents the monodromy φ_u, possibly after passing to the *stable quotient* of the map $(f_u)_*$.

Knowing that each monodromy φ_u is fully irreducible, it now makes sense to discuss the attracting $\mathbb{R}$-tree $T^u_+ = T_+(\varphi_u)$ for each surjective $u\colon G \to \mathbb{Z}$ with $u \in \Sigma_0$. We describe a space that relates them now, defined by Dowdall–Kapovich–Leininger [DKL26]. Let $\tilde{X}$ be the universal cover of the mapping torus X with lifted semiflow $\tilde{\psi}$. For $x, y \in \tilde{X}$, declare $x \sim_{\tilde{\psi}} y$ if there exist $t, s \geq 0$ so that $\tilde{\psi}_t(x) = \tilde{\psi}_s(y)$. In special cases, the universal real tree is $\tilde{X}/\sim_{\tilde{\psi}}$; in general $\sim_{\tilde{\psi}}$ may not be closed as a subset of $\tilde{X} \times \tilde{X}$, so $X/\sim_{\tilde{\psi}}$ may not be Hausdorff. The non-Hausdorff points are images of endpoints of *(pre)-periodic Nielsen paths* of the defining train track map f. Regardless, $\tilde{X}/\sim_{\tilde{\psi}}$ has a Hausdorff quotient: let $\sim$ be obtained by first closing $\sim_{\tilde{\psi}}$ in $\tilde{X} \times \tilde{X}$, then taking the transitive closure of the resulting relation. The *universal real tree* $(T_X, \mathcal{T}_{\text{univ}})$ is the quotient $X/\sim$, see Figure 2.

Theorem 3.1 ([DKL26])**.** *Let $u\colon G \to \mathbb{Z}$ be surjective with $u \in \Sigma_0$; let $(T_u, \mathcal{T}_{\mathbb{R}})$ be the attracting $\mathbb{R}$-tree of its monodromy φ_u. There is a G-equivariant homeomorphism $(T_X, \mathcal{T}_{univ}) \to (T_u, \mathcal{T}_{int})$.*

Otherwise stated, after passing to the intervals topology, the attracting $\mathbb{R}$-tree is canonically associated to a component of the BNS invariant. For our purposes, the observers' topology is more central than the intervals topology, so we record the following corollary.

Corollary 3.2. *Let $u, u'\colon G \to \mathbb{Z}$ be in the same component of $\Sigma(G)$, whose monodromies have attracting $\mathbb{R}$-trees $(T_u, \mathcal{T}^u_{\mathbb{R}}), (T_{u'}, \mathcal{T}^{u'}_{\mathbb{R}})$, respectively. There is a G-equivariant homeomorphism $(T_u, \mathcal{T}^u_{obs}) \to (T_{u'}, \mathcal{T}^{u'}_{obs})$.*

Proof. By Theorem 3.1, the universal real tree $(T_X, \mathcal{T}_{\text{univ}})$ is a real tree that is G-equivariantly homeomorphic to $(T_u, \mathcal{T}^u_{\text{int}})$. The intervals topology and the real tree topology on T_u,

have the same arcs, thus the same directions. Thus, passing to the observers' topology from $(T_u, \mathcal{T}^u_{\mathbb{R}})$ or $(T_u, \mathcal{T}^u_{\text{int}})$ result in the same space $(T_u, \mathcal{T}^u_{\text{obs}})$. In particular, there is a G-equivariant homeomorphism $(T_X, \mathcal{T}_{\text{obs}}) \to (T_u, T^u_{\text{obs}})$. □

4. Algebraic laminations and the map $\mathcal{Q}$

4.1. **Hyperbolic groups.** In this section we give a brief overview of hyperbolic spaces and groups [Gro87]; here we follow [BH99]. Let (X, d) be a metric space. The *Gromov product* $(x \mid y)_z$ is the quantity

$$(x \mid y)_z = \frac{1}{2}(d(x, z) + d(y, z) - d(x, y)).$$

For a fixed $\delta \geq 0$, we say that X is *δ-hyperbolic* if for any $x, y, z, P \in X$, we have that

$$(x \mid y)_P \geq \min\{(x \mid z)_P, (z \mid y)_P\} - \delta.$$

The space X is *hyperbolic* if X is δ-hyperbolic for some $\delta \geq 0$. We note that $\mathbb{R}$-trees are precisely the 0-hyperbolic geodesic metric spaces.

Definition 4.1. A function $f\colon X \to Y$ of metric spaces is a *K-quasi-isometric embedding* if for any $x, y \in X$,

$$\frac{1}{K} d_Y(f(x), f(y)) - K \leq d_X(x, y) \leq K d_Y(f(x), f(y)) + K.$$

If X is a subinterval of $\mathbb{R}$ or a set of consecutive integers, we call a K-quasi-isometric embedding a *K-quasi-geodesic*. If f is a K-quasi-isometric embedding and each point of Y is bounded distance to the image of f, we say f is a *quasi-isometry*.

The following result will be useful later and motivates the definition of the Gromov product. It follows directly from the Morse lemma, see [BH99, Theorem H.1.7 and p. 410].

Lemma 4.2. *There is a constant M depending only on K and δ such that if X is a geodesic δ-hyperbolic space and γ is a K-quasi-geodesic between x and y, we have that*

$$d(P, \gamma) - M \leq (x \mid y)_P \leq d(P, \gamma) + M$$

for any $P \in X$.

Hyperbolicity of a geodesic metric space is a quasi-isometry invariant, although the associated δ may change. We say a finitely generated group G is *hyperbolic* if some (any) Cayley graph $\text{Cay}(G)$ of G is hyperbolic.

Given a sequence $x_n \in X$, we say that x_n *converges at infinity* if $(x_i, x_j)_P \to \infty$ as $i, j \to \infty$. Sequences x_n and y_n are equivalent if $(x_i, y_j)_P \to \infty$ as $i, j \to \infty$. The *Gromov boundary* $\partial_\infty X$ is the set of equivalence classes of sequences that converge at infinity. When X is a proper geodesic space, $\partial_\infty X$ is identified with the space of asymptote classes of quasi-geodesic rays.

Example 4.3. We give a few examples of hyperbolic groups and their boundaries.

- The free group F_N is hyperbolic and for $N \geq 2$, $\partial_\infty F_N$ is a Cantor set.
- The fundamental group of a closed hyperbolic n-manifold is hyperbolic and has Gromov boundary S^{n-1}.
- A free-by-cyclic group G is hyperbolic if and only if its monodromy is atoroidal [Bri00]. When G has irreducible monodromy, $\partial_\infty G$ is the Menger curve [KK00].

The Gromov product extends to $\partial_\infty X$ by the following formula

$$(x \mid y)_P = \sup_{x_n \to x, y_n \to y} \liminf_{i,j} (x_i \mid y_j)_P.$$

If X is both proper and geodesic, then $(x \mid y)_P = \infty$ if and only if $x = y$, and a sequence $x_n \in \partial_\infty X$ converges to $y \in \partial_\infty X$ if and only if the sequence $(x_n \mid y)_P$ goes to infinity with n, see [BH99, Remarks H.3.17].

4.2. **Bounded backtracking and the map $\mathcal{Q}$.** In this section, all trees will be equipped with an isometric F_N-action and are in $\overline{CV}_N$. Our goal is to describe a certain equivariant map $\mathcal{Q}\colon \partial_\infty F_N \to \widehat{T}$. The map $\mathcal{Q}$ was first defined by Levitt–Lustig [LL03], building on ideas from [GJLL98]. See [CHL07; CHL08b; LL08; KL15] for other discussion of and uses of $\mathcal{Q}$.

We first describe a *bounded backtracking property* of $\mathbb{R}$-trees: let $f_0\colon T_0 \to T$ be an equivariant map from a simplicial F_N-tree to some $\mathbb{R}$-tree T. We say that f_0 has *bounded backtracking* if there exists a *bounded backtracking constant* $BBT(f_0) > 0$ such that, for any arc $[P, Q]$ in T_0, $f_0([P, Q])$ is in the $BBT(f_0)$-neighborhood of $[f_0(P), f_0(Q)]$. We also define the *volume* of T_0/F_N, denoted $\mathrm{vol}(T_0/F_N)$, to be the sum of the edge lengths of T_0/F_N.

Lemma 4.4 ([BFH97], Lemma 3.1)**.** *Let T_0 be a metric simplicial tree, T an $\mathbb{R}$-tree, and $f_0\colon T_0 \to T$ an equivariant map. The map f_0 has bounded backtracking with bounded backtracking constant $BBT(f_0) \leq \mathrm{vol}(T_0/F_N)\mathrm{Lip}(f_0)$, where $\mathrm{Lip}(f_0)$ is the Lipschitz constant of f_0.*

Lemma 4.5 ([LL03], Corollary 2.3)**.** *For any $\mathbb{R}$-tree T there is a sequence of equivariant maps $f_n\colon T_n \to T$ with $BBT(f_n) \to 0$ as $n \to \infty$.*

As in the prior section, given an $\mathbb{R}$-tree T, let $\overline{T}$ be its metric completion and $\widehat{T} = \overline{T} \cup \partial_\infty T$ be $\overline{T}$ together with its Gromov boundary.

We now introduce a bit of terminology before defining $\mathcal{Q}$. Fix an $\mathbb{R}$-tree T, and let T_0 be a simplicial tree with an equivariant map $f_0\colon T_0 \to T$. The F_N action on T_0 identifies $\partial_\infty F_N$ with $\partial_\infty T_0$. Pick some basepoint $P_0 \in T_0$; for any $x \in \partial_\infty F_N$, let $\rho_x\colon [0, \infty)$ be the geodesic ray based at P_0 representing x. We say x is *T-unbounded* if the composition $f_0 \circ \rho_x$ has unbounded image in T_0. In this case, $f_0(\rho)$ is bounded distance to a geodesic ray in T by bounded backtracking, We note that this notion does not depend on f_0 or P_0. Likewise, x is *T-bounded* if $f_0 \circ \rho_x$ has bounded image; in this case, after an initial segment, the image of $f_0 \circ \rho_x$ stays in a $2BBT(f_0)$-diameter subset of $T \subseteq \overline{T}$. We can now define the map $\mathcal{Q}$.

Definition 4.6 ([LL03])**.** For any $\mathbb{R}$-tree T there is an equivariant function $\mathcal{Q}\colon \partial_\infty F_N \to \widehat{T}$ defined in the following manner. If x is T-unbounded, for any equivariant map $f_0\colon T_0 \to T$ and any ray $\rho_x\colon [0, \infty) \to T_0$ representing x, $f_0 \circ \rho([0, \infty))$ is bounded distance to a geodesic ray representing a point $\mathcal{Q}(x) \in \partial_\infty T \subset \widehat{T}$. If x is T-bounded, we define $\mathcal{Q}(x)$ to be the unique point such that, for any equivariant map $f_0\colon T_0 \to T$ and any ray $\rho_x\colon [0, \infty) \to T_0$ representing x, $\mathcal{Q}(x)$ is in the closure of the image of ρ_x within $\overline{T}$.

The map $\mathcal{Q}$ is well-defined for T-bounded $x \in \partial_\infty F_N$ by Lemma 4.5. We summarize some properties of $\mathcal{Q}$ in the following proposition.

Proposition 4.7. *Let T, $\widehat{T}$ be as above.*

(1) $\mathcal{Q}$ is surjective;

(2) If $\widehat{T}$ has the observers' topology, $\mathcal{Q}$ is continuous. In fact, $\mathcal{Q}$ is a quotient map.

Proof. (1) is [LL03, Corollary 3.5]; see also [CHL07, Proposition 2.4]. (2) is [CHL07, Proposition 2.3 and Corollary 2.6]. □

4.3. **Algebraic laminations.** For a free group F_N, consider the double boundary $\partial_\infty F_N \times \partial_\infty F_N \setminus \Delta$, where Δ is the diagonal; we call the involution swapping the factors a *flip*. An *algebraic lamination* is a non-empty, closed, flip-invariant subset $L \subseteq \partial_\infty F_N \times \partial_\infty F_N \setminus \Delta$ [CHL08a]. If $(x, y) \in L$, we refer to (x, y) as a *leaf* of L; x and y are *half-leaves* of L. We denote by $L^1 \subseteq \partial_\infty F_N$ the set of half-leaves of L. Given $x \in \partial_\infty F_N$, there is a minimal lamination $L(x)$ associated to x most easily defined in terms of (bi)infinite words in a basis of F_N, see [CHL08b; KL15]. If $L(x) \subseteq L$, we say x is an *end* of L and denote the set of ends of L by $\mathrm{Ends}(L)$. We note that the inclusion $L^1 \subseteq \mathrm{Ends}(L)$ may be strict. We remark that algebraic laminations can be analagously defined for hyperbolic groups.

The *diagonal closure* of an algebraic lamination L is the lamination

$$\mathrm{diag}(L) = \{(x, y) \in \partial_\infty F_N \times \partial_\infty F_N \setminus \Delta \mid \exists\, x = x_0, \dots, x_n = y \text{ such that } (x_i, x_{i+1}) \in L\}.$$

Bestvina–Feighn–Handel showed that a fully irreducible $\varphi \in \mathrm{Out}(F_N)$ preserves a pair of algebraic laminations $L_{\mathrm{BFH}}(\varphi)$ and $L_{\mathrm{BFH}}(\varphi^{-1})$; these laminations are not necessarily diagonally closed, in that the containment $L_{\mathrm{BFH}}(\varphi) \subseteq \mathrm{diag}(L_{\mathrm{BFH}}(\varphi))$ may be strict.

4.4. **The dual lamination to an $\mathbb{R}$-tree.** For an isometric action of F_N on an $\mathbb{R}$-tree T, for each $w \in F_N$, its *translation length* is $\mathrm{tr}_T(w) = \inf_{x \in T} d(x, wx)$; in this setting it is always attained and the set $\mathrm{axis}_T(w) = \{x \in T \mid d(x, wx) = \mathrm{tr}_T(w)\}$ is called the *axis* of w and is isometric to $\mathbb{R}$. For $w \in F_N$, denote by $w^\pm$ its attracting and repelling fixed points in $\partial_\infty F_N$. We can now make the following definition, which is a rephrasing of the one due to Coulbois–Hilion–Lustig [CHL08b].

Definition 4.8. Let T be an $\mathbb{R}$-tree with a very small F_N-action. The *dual lamination* $L(T) \subseteq \partial_\infty F_N \times \partial_\infty F_N \setminus \Delta$ is the set of (x, y) such that there exists a sequence $w_n \in F_N$ with $w_n^- \to x$, $w_n^+ \to y$, and $\mathrm{tr}_T(w_n) \to 0$ as $n \to \infty$.

If $L(T) \neq \varnothing$, $L(T)$ is an algebraic lamination. For the purposes of this paper, the reader is free to assume that T is the attracting or repelling tree of a fully irreducible $\varphi \in \mathrm{Out}(F_N)$. The following proposition of Coulbois–Hilion–Lustig is one of the key tools we use to understand the map $\mathcal{Q}$.

Proposition 4.9 ([CHL08b], Proposition 8.5)**.** *For $(x, y) \in \partial_\infty F_N \times \partial_\infty F_N \setminus \Delta$, $\mathcal{Q}(x) = \mathcal{Q}(y)$ if and only if $(x, y) \in L(T)$.*

We note the following, as observed by Kapovich–Lustig in [KL15]

$$L^1(T) \subseteq \mathcal{Q}^{-1}(\overline{T}) \subseteq \mathrm{Ends}(T)$$

When we construct $\mathfrak{C}$, it will be important for us that $\mathcal{Q}^{-1}(\overline{T}_+)$ and $\mathcal{Q}^{-1}(\overline{T}_-)$ are disjoint. This follows from another observation of Kapovich–Lustig in [KL15], although the work of showing this was mostly carried out in [KL14].

Proposition 4.10 ([KL15], Remark 3.15 and Proposition 3.22)**.** *If φ is fully irreducible and $T_\pm = T_\pm(\varphi)$, $L(T_+)$ and $L(T_-)$ share no common half-leaves. Thus* $\mathrm{Ends}(L(T_+))$ *and* $\mathrm{Ends}(L(T_-))$ *are disjoint and $\mathcal{Q}^{-1}(\overline{T}_+) \cap \mathcal{Q}^{-1}(\overline{T}_-) = \varnothing$.*

4.5. **Cannon–Thurston maps and ending laminations.** Given a short exact sequence of groups

$$1 \to H \to G \to Q \to 1$$

where H and G are hyperbolic, Mj showed the the inclusion of Cayley graphs $\mathrm{Cay}(H) \to \mathrm{Cay}(G)$ extends to a continuous map $i\colon \mathrm{Cay}(H) \cup \partial_\infty H \to \mathrm{Cay}(G) \cup \partial_\infty G$ [Mit98]. We will be concerned with the restriction $i\colon \partial_\infty H \to \partial_\infty G$, which we refer to as a Cannon–Thurston map; Mj's theorem generalized a result of Cannon and Thurston for hyperbolic 3-manifolds fibering over S^1 [CT07].

When H is not virtually $\mathbb{Z}$, Mosher showed that Q is hyperbolic as well [Mos96]. In this case, Mj associated to each point $z \in \partial_\infty Q$ an algebraic *ending lamination* $\Lambda_z \subseteq \partial_\infty H \times \partial_\infty H \setminus \Delta$ [Mit97]. Mj proved that these laminations precisely determine the failure of injectivity for i: for any distinct $x, y \in \partial_\infty H$, $i(x) = i(y)$ if and only if there is some $z \in \partial_\infty Q$ such that $(x, y) \in \Lambda_z(Q)$. This again generalized the 3-manifold setting.

When $Q \cong \mathbb{Z}$, $\partial_\infty Q = \{\pm\infty\}$. When H is free, it is natural to ask how these laminations interact with the other laminations we have discussed. Kapovich and Lustig proved the following theorem [KL14; KL15], see also [Fie20]:

Theorem 4.11 (Kapovich–Lustig)**.** *Let $\varphi \in \mathrm{Out}(F_N)$ be fully irreducible and atoroidal. The following are equal*

$$L(T_+(\varphi)) = \Lambda_{-\infty} = \mathrm{diag}(L_{BFH}(\varphi^{-1})).$$

Corollary 4.12. *The Cannon–Thurston map $i\colon \partial_\infty F_N \to \partial_\infty G$ factors through maps*

$$\partial_\infty F_N \xrightarrow{\mathcal{Q}_\pm} \widehat{T}_\pm(\varphi) \xrightarrow{\mathcal{R}_\pm} \partial_\infty G,$$

where $\mathcal{Q}_\pm$ is the equivariant map of Definition 4.6 and $\mathcal{R}_\pm$ is injective and continuous with respect to the observers' topology on $\widehat{T}_\pm(\varphi)$.

5. The universal Cantor set

In this section we construct a G-action on a Cantor set $\mathfrak{C}$ associated to a component of $\Sigma_s(G)$. For a rational $u \in \Sigma_s(G)$, we will first define a G action on a set $\mathfrak{C}_u$ defined solely in terms of $T_+(\varphi_u)$ and $T_-(\varphi_u)$. We will show that $\mathfrak{C}_u$ is G-equivariantly identified to $\partial_\infty \ker u = \partial_\infty F_N$. To show that $\mathfrak{C}_u$ does not depend on the choice of u within Σ_0, we will pass our construction to the end spaces of the trees $T_\pm$ via Proposition 2.11 and use Dowdall–Kapovich–Leininger's universal real tree to associate $\mathfrak{C} \cong \mathfrak{C}_u$ canonically to a component of $\Sigma_s(G)$. The fact that $\mathfrak{C}$ is a Cantor set will follow from the identification with $\partial_\infty \ker u$.

Let $\varphi \in \mathrm{Out}(F_N)$, $T_\pm = T_\pm(\varphi)$ be attracting and repelling trees, and $L(T_\pm) \subseteq \partial_\infty F_N \times \partial_\infty F_N \setminus \Delta$ be its dual lamination. Denote by $G = F_N \rtimes_\varphi \mathbb{Z}$ the associated free-by-cyclic group and $u\colon G \to \mathbb{Z}$ the quotient by F_N, with associated monodromy $\varphi = \varphi_u$. This defines a component Σ_0 of $\Sigma_s(G)$. Recall that G acts on the trees $T_\pm$ by homotheties.

We call a point $z \in \widehat{T}_+$ a *laminar point* if there is a leaf (x, y) of $L(T_+)$ such that $\mathcal{Q}_+(x) = \mathcal{Q}_+(y) = z$ and we will let T_+^{lam} be the union of T_+ with the set of laminar points; note that $T_+^{\mathrm{lam}} \subseteq \overline{T}_+$. We note that $z \in \widehat{T}_+$ is a laminar point if and only if $\mathcal{R}_-^{-1}(\mathcal{R}_+(z)) \subseteq (\widehat{T}_- \setminus \overline{T})$ has multiple points. We make a similar definition for $\widehat{T}_-$. The set of laminar points is G-invariant by G-equivariance of $\mathcal{R}_\pm$.

Definition 5.1. Giving $\widehat{T}_\pm$ the observers' topology, we define

$$\mathfrak{C}_u = (\widehat{T}_+ \setminus T_+^{\mathrm{lam}}) \sqcup (\widehat{T}_- \setminus T_-^{\mathrm{lam}}) / \sim,$$

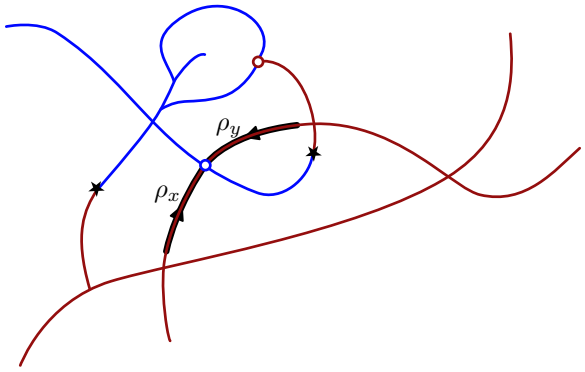


FIGURE 3. A cartoon of $\mathcal{R}_+(T_+)$ (in red) and $\mathcal{R}_-(T_-)$ (in blue) within $\partial_\infty G$. The open blue dot is a laminar point of $\overline{T}_-$. Despite the rays ρ_x, ρ_y converging to the same point of $\partial_\infty G$, they will be distinct points of the universal Cantor set $\mathfrak{C}$. The open red dot is a laminar point in $\overline{T}_+ \setminus T_+$. The ends of $T_\pm$ converging to the stars will be identified in $\mathfrak{C}$.

where $\sim$ identifies $x \in \widehat{T}_+$ with $y \in \widehat{T}_-$ if $\mathcal{R}_+(x) = \mathcal{R}_-(y) \in \partial_\infty G$. We will write $Y_\pm$ for $\widehat{T}_\pm \setminus T_\pm^{\text{lam}}$, $Y = Y_+ \sqcup T_-$, and $p\colon Y \to \mathfrak{C}_u$ the quotient map.

By the definition of $\sim$, $\mathcal{R}_+ \sqcup \mathcal{R}_-$ descends to a continuous map $\mathcal{R}\colon \mathfrak{C}_u \to \partial_\infty G$. We note that the exclusion of laminar points ensures that $\sim$ is transitive: no identification is made between distinct points of $\widehat{T}_+$ or $\widehat{T}_-$, see Figure 3. We remark that in the surface setting, $T_\pm^{\text{lam}} = T_\pm$; we cannot say for certain whether this holds in general.

Remark 5.2. The identification $\sim$ of Definition 5.1 should be compared with the 'bridge correspondence' used to identify the universal circles of the zippers in [CL26b].

We will now describe the topology of $\mathfrak{C}_u$. We note that, since $\mathcal{Q}_\pm$ are quotient maps by Proposition 4.7 and by 4.9 are injective on $\partial_\infty F_N \setminus \mathcal{Q}_\pm^{-1}(T_\pm^{\text{lam}})$, the restricted maps $\mathcal{Q}_\pm\colon \partial_\infty F_N \setminus \mathcal{Q}_\pm^{-1}(T_\pm^{\text{lam}}) \to \widehat{T}_\pm \setminus T_\pm^{\text{lam}}$ are G-equivariant homeomorphisms. Thus, there are induced maps $q_\pm\colon \widehat{T}_\mp \setminus T_\mp^{\text{lam}} \to \widehat{T}_\pm$ defined by $q_\pm = \mathcal{Q}_+ \circ \mathcal{Q}_-^{-1}$. We record the following basic properties.

Lemma 5.3. *Let $y_+ \in Y_+$ and $y_- \in Y_-$. The following are equivalent*

(1) $q_-(y_+) = y_-$;
(2) $q_+(y_-) = y_+$;
(3) $p(y_+) = p(y_-)$;
(4) There is a point $x \in \partial_\infty F_N$ such that $\mathcal{Q}_+(x) = y_+$ and $\mathcal{Q}_-(x) = y_-$.

Proof. We will prove that each of (1), (2), and (3) are equivalent to (4).

Suppose (1), since $y_+ \in Y_+ = \widehat{T}_+ \setminus T_+^{\text{lam}}$, there is a well-defined point $x = \mathcal{Q}_+^{-1}(y_+)$ and

$$y_- = q_-(y_+) = \mathcal{Q}_-(x).$$

Thus (1) implies (4), similarly (2) implies (4).

Assuming (4), $\mathcal{Q}_\pm(x) = y_\pm$. By the assumption that $y_\pm \in Y_\pm$, x is the unique point of $\partial_\infty F_N$ satisfying this for either y_+ or y_-. Thus $q_\pm(y_\mp) = y_\pm$ and (4) implies (1) and (2).

Assume (3), there are unique points $x_+, x_- \in \partial_\infty F_N$ such that $\mathcal{Q}_\pm(x_\pm) = y_\pm$. Corollary 4.12 yields

$$i(x_+) = \mathcal{R}_+(\mathcal{Q}_+(x_+)) = \mathcal{R}_+(y_+) = \mathcal{R}(p(y_+)) = \mathcal{R}(p(y_-)) = \mathcal{R}_-(y_-) = \mathcal{R}_-(\mathcal{Q}_-(x_-)) = i(x_-).$$

We notice that since $\mathcal{Q}_+(x_+) \notin T_+^{\mathrm{lam}}$, $x_+ \notin L^1(T_+)$; likewise, $x_- \notin L^1(T_-)$. Thus $i(x_+) = i(x_-)$ but (x_+, x_-) is not a leaf of $L(T_+)$ or $L(T_-)$, so $x_+ = x_-$.

Conversely, assume (4). Again by Corollary 4.12, we have that

$$\mathcal{R}_+(y_+) = \mathcal{R}_+(\mathcal{Q}_+(x)) = i(x) = \mathcal{R}_-(\mathcal{Q}_-(x)) = \mathcal{R}_-(y_-).$$

This is precisely the statement that $y_+ \sim y_-$, so $p(y_+) = p(y_-)$. □

Corollary 5.4. *Let $p\colon Y \to \mathfrak{C}_u$ be the usual projection and suppose $A_\pm$ are open subsets of $Y_\pm$, consider $A = A_+ \sqcup A_- \subseteq Y_+ \sqcup Y_- = Y$. The set A satifies $A = p^{-1}(p(A))$ if and only if $q_+(A_-) \cap Y_+ \subseteq A_+$ and $q_-(A_+) \cap Y_- \subseteq A_-$.*

Proof. Suppose $q_\pm(A_\mp) \cap Y_\pm \subseteq A_\pm$. If $y \in p^{-1}(p(A))$, there is some $x \in A$ with $p(x) = p(y)$. Without loss of generality, assume that $x \in A_+$. If $y \in Y_+$ as well, then $y = x$ as p is injective on Y_+. If $y \in Y_-$ then by Lemma 5.3 $y = q_-(x) \in A_-$. On the other hand, if for instance $x_+ \in A_+$ and $q_-(x_+) \in Y_- \setminus A_-$, then $p(x_+) = p(x_-)$ and so $A \neq p^{-1}(p(A))$. □

We now define a map $h_u\colon \partial_\infty F_N \to \mathfrak{C}_u$, which we will later show is a homeomorphism. Let $x \in \partial_\infty F_N$, by Proposition 4.10, either $\mathcal{Q}_+(x) \notin T_+^{\mathrm{lam}}$ or $\mathcal{Q}_-(x) \notin T_-^{\mathrm{lam}}$. If exactly one of these holds, say for $\mathcal{Q}_+$, we define $h_u(x) = \mathcal{Q}_+(x)$ as a point of $\mathfrak{C}_u$. Otherwise, if $\mathcal{Q}_\pm(x) \in \widehat{T}_\pm \setminus (T_\pm \cup T_\pm^{\mathrm{lam}})$ for both trees $T_\pm$, then $\mathcal{R}_+(\mathcal{Q}_+(x)) = i(x) = \mathcal{R}_-(\mathcal{Q}_-(x))$ [KL15, Proposition 4.8] and we declare $h_u(x)$ to be the equivalence class $\{\mathcal{Q}_+(x), \mathcal{Q}_-(x)\}$.

Proposition 5.5. *The map $h_u\colon \partial_\infty F_N \to \mathfrak{C}_u$ is a G-equivariant open bijection.*

Proof. First, we show that h_u is injective. Let x, y be distinct elements of $\partial_\infty F_N$. If $i(x) \neq i(y)$, then $h_u(x) \neq h_u(y)$. If $i(x) = i(y)$, then (x, y) is a leaf of $L(T_+)$ or of $L(T_-)$ by Theorem 4.11, for now say $L(T_+)$. By Proposition 4.10, $\mathcal{Q}_-(x) \neq \mathcal{Q}_-(y)$. Thus, $h_u(x) \neq h_u(y)$.

Let $z \in \mathfrak{C}_u$. If there is a $y_+ \in Y_+$ such that $p(y_+) = z$, there is some $x \in \partial_\infty F_N$ such that $\mathcal{Q}_+(x) = y_+$ by Proposition 4.7 and so $h_u(x) = z$. Otherwise, there is a $y_- \in Y_-$ with $p(y_-) = z$ and we argue similarly. Thus, h_u is surjective.

We will show that h_u is an open map. Let $U \subseteq \partial_\infty F_N$ be open. Consider $U_\pm = \mathcal{Q}_\pm(U) \cap Y_\pm$. Note that the restricted maps $\mathcal{Q}_\pm\colon \partial_\infty F_N \setminus \mathcal{Q}_\pm^{-1}(T_\pm^{\mathrm{lam}}) \to Y_\pm$ are homeomorphisms by Proposition 4.7, as this restriction is an injective quotient map. Since $U \setminus \mathcal{Q}_\pm^{-1}(T_\pm^{\mathrm{lam}})$ is open in $\partial_\infty F_N \setminus \mathcal{Q}_\pm^{-1}(T_\pm^{\mathrm{lam}})$, $U_\pm$ is open in $Y_\pm = \widehat{T}_\pm \setminus T_\pm^{\mathrm{lam}}$. Notice that $q_\pm(U_\mp) \cap Y_\pm \subseteq U_\pm$, thus $U_+ \sqcup U_-$ is evenly covered by Corollary 5.4. In particular, $h_u(U) = p(U_+ \sqcup U_-)$ is open in $\mathfrak{C}_u$.

Finally, h_u is G-equivariant as $\mathcal{Q}_\pm$ are G-equivariant. □

Lemma 5.6. *The space $\mathfrak{C}_u$ is sequentially compact.*

Proof. Let $y_n \in Y$ be a sequence. We wish to show that $p(y_n)$ has a convergent subsequence in $\mathfrak{C}_u$. Infinitely many y_n are in Y_+ or Y_-, suppose Y_+. Pass to a subsequence consisting of points in $Y_+ \subseteq \widehat{T}_+$. By Lemma 2.9, we may pass to a further subsequence so that y_n converges to a point $y_+ \in \widehat{T}_+$. If $y \in Y_+$, then $p(y_n)$ converges to $p(y_+)$. Otherwise, consider

the sequence $q_-(y_n)$ in Y_-. Passing to a subsequence, $q_-(y_n)$ converges to a point $y_- \in \widehat{T}_-$. For each n, there is some $x_n \in \partial_\infty F_N$ such that $\mathcal{Q}_+(x_n) = y_n$ and $\mathcal{Q}_-(y_n) = q_-(y_n)$. Thus,

$$\mathcal{R}_+(y_n) = \mathcal{R}_+(\mathcal{Q}_+(x_n)) = i(x_n) = \mathcal{R}_-(\mathcal{Q}_-(x_n)) = \mathcal{R}_-(q_-(y_n)).$$

Taking limits, the above yields $\mathcal{R}_+(y_+) = \mathcal{R}_-(y_-)$. Since $y_+ \in T_+^{\text{lam}} \subseteq \overline{T}_+$, $y_- \notin \mathcal{R}_-(\overline{T}_-)$, so $y_- \in Y_-$ and $p(y_n)$ converges to $p(y_-)$. □

The following is a version of the more standard theorem for continuous bijections from compact spaces to Hausdorff spaces, found for instance on nLab [nLa26].

Lemma 5.7. *Let A be sequentially compact, and B metrizable. Any continuous bijection $f\colon A \to B$ is a homeomorphism.*

Proposition 5.8. *The map $h_u\colon \partial_\infty F_N \to \mathfrak{C}_u$ is a G-equivariant homeomorphism.*

Proof. By Proposition 5.5, $h_u^{-1}\colon \mathfrak{C}_u \to \partial_\infty F_N$ is a continuous bijection. The boundary $\partial_\infty F_N$ is a Cantor set, which is metrizable. By Lemma 5.6, $\mathfrak{C}_u$ is sequentially compact and we may apply the Lemma 5.7 to see that h_u^{-1} is a homeomorphism. □

5.1. **Across the mapping torus.** In this section we return to the mapping torus perspective and prove Theorem A.

Given an expanding irreducible train track map $f\colon \Gamma \to \Gamma$ representing φ, define a pseudo-metric d_k on $\tilde{\Gamma}$ by

$$d_k(x,y) = \frac{d(\tilde{f}^k(x), \tilde{f}^k(y)}{\lambda^k},$$

where λ is the stretch factor of f. We denote $\tilde{\Gamma}_k$ to be the quotient of $\tilde{\Gamma}$ by equating points of d_k-distance zero, we write $p_k\colon \tilde{\Gamma} \to \tilde{\Gamma}_k$. Each $\tilde{\Gamma}_k$ is simplicial, and admits a 1-Lipschitz map $f_k\colon \tilde{\Gamma}_k \to T_+$, where T_+ is given the normalization of Section 2.1. If Γ has volume one, $\tilde{\Gamma}_k/F_N$ has volume λ^{-k}, so the bounded backtracking constant $BBT(f_k) \leq \lambda^{-k}$ by Lemma 4.4. We note that $BBT(f_k)$ depends on the metric on T_+, but not on the scalar representative of the metric on $\tilde{\Gamma}_k$.

Lemma 5.9. *Suppose $z \in \partial_\infty F_N$ is such that $\mathcal{Q}_+(z) \in T$. There is a sequence of points $z_k \in \tilde{\Gamma}$ such that $z_k \to z$ and $f_0(z_k)$ metrically converges to $\mathcal{Q}_+(z)$ in T_+.*

Proof. Let $x \in \tilde{\Gamma}_0$ have $f_0(x) = \mathcal{Q}_+(z)$ and define $\rho_0 : [0,\infty) \to \tilde{\Gamma}_0$ to be the tight ray converging to z such that $\rho_0(0) = x$. For each k, let ρ_k be the tightening of $p_k \circ \rho_0$.

For each k, let $z_k' \in \rho_k \subseteq \tilde{\Gamma}_k$ have distance at least k from $\rho_k(0)$. Since ρ_k is a tightening of $p_k \circ \rho_0$, there is a point $z_k \in \rho_0$ with $p_k(z_k) = z_k'$. The map p_k is 1-Lipschitz, so $d_{\tilde{\Gamma}_0}(\rho_0(0), z_k) \geq k$. Thus $z_k \to z$ as $k \to \infty$.

For each k, $BBT(f_k) \leq \lambda^{-k}$. Fix $\varepsilon > 0$ and let $k > \log_\lambda(\varepsilon)$. We have that $f_k \circ \rho_k([0,\infty))$ is in the $BBT(f_k) = \lambda^{-k}$-neighborhood of $\mathcal{Q}(z)$, so

$$d(f_0(z_k), \mathcal{Q}_+(z)) = d(f_k(z_k'), \mathcal{Q}_+(z)) \leq BBT(f_k) \leq \lambda^{-k} < \varepsilon.$$

This proves the result. □

Now, as in Section 2, let X be the mapping torus of f with suspension semiflow ψ and 'fibration' $\eta : X \to S^1$. Let $\tilde{X}$ be the universal cover with lifted semiflow $\tilde{\psi}$ and fibration $\tilde{\eta} : \tilde{X} \to \mathbb{R}$. We note that $\tilde{\eta}(\tilde{\psi}_t(x)) = \tilde{\eta}(x) + t$ for all $x \in \tilde{X}$ and $t \geq 0$. The universal cover $\tilde{\Gamma}$ of Γ includes into $\tilde{X}$ as $\tilde{\eta}^{-1}(\{0\})$, we denote $\tilde{\Gamma}_k = \tilde{\eta}^{-1}(\{k\})$ for all $k \geq 0$ and note that the map $\tilde{\psi}_k\colon \tilde{\Gamma} \to \tilde{\Gamma}_k$ agrees with p_k.

Define a map $\Psi_0^+ \colon \tilde{\Gamma} \to \partial_\infty X$ by sending x to the forward endpoint of the semiflow line $\psi_t(x)$. This map Ψ_0^+ is constant on fibers of f_0: either distinct points will flow together eventually or they are the endpoints of (pre)-periodic Nielsen paths, or concatenations thereof, which have asymptotic flow lines. Thus Ψ_0^+ descends to a map $\Psi^+ \colon T_+ \to \partial_\infty G$ that is continuous with respect to the metric topology on T_+.

Proposition 5.10. *For all $x \in T_+$, $\Psi^+(x) = \mathcal{R}_+(x)$.*

Proof. Let $z \in \partial_\infty F_N$ have $\mathcal{Q}_+(z) = x$. Let $z_n \in \tilde{\Gamma}_0$ be a sequence such that $z_n \to z$ and $f_0(z_n)$ metrically converges to $\mathcal{Q}_+(z)$, as in Lemma 5.9. The map Ψ^+ is continuous when T_+ has the metric topology, so

$$\lim_{n\to\infty} \Psi^+(f_0(z_n)) = \Psi^+(\mathcal{Q}_+(z)) = \Psi^+(x).$$

To prove the proposition, we will show that the limit above also converges to $\mathcal{R}_+(x)$. Fix a geodesic metric on X, there is some $K \geq 1$ large enough so that $\tilde{\eta}$ is K-Lipschitz and that for all $y \in \tilde{X}$ the function $t \mapsto \tilde{\psi}_t(y)$ is a K-quasi-geodesic.

Pick a basepoint $P \in \tilde{X}$. For each n, consider the flow line $\tilde{\psi}_t(z_n)$. Since $\tilde{\eta}$ is K-Lipschitz, we have

$$d(\tilde{\psi}_t(z_n), P) \geq |\tilde{\eta}(\tilde{\psi}_t(z_n)) - \tilde{\eta}(P)|/K = t/K.$$

Now, since semiflow lines are quasi-geodesics, we additionally have

$$d(z_n, P) \leq d(z_n, \tilde{\psi}_t(z_n)) + d(\tilde{\psi}_t(z_n), P) \leq Kt + K^2 + d(\tilde{\psi}_t(z_n), P)$$

which implies

$$d(\tilde{\psi}_t(z_n), P) \geq d(z_n, P) - Kt - K^2.$$

Together, we have that

$$d(\tilde{\psi}_t(z_n), P) \geq \inf_{t\geq 0} \max\left\{d(z_n, P) - Kt - K^2, t/K\right\}$$

We call the quantity at the right of the above r_n, note that $r_n \to \infty$ as $n \to \infty$, since $d(z_n, P) \to \infty$ as $n \to \infty$. Now we note that, since z_n defines the point $i(z) \in \partial_\infty G$, $(z_n \mid z_j)_P \to \infty$ as $n, j \to \infty$. Otherwise stated, the sequence $a_n = \liminf_{j\to\infty}(z_n \mid z_j)_P$ goes to infinity as $n \to \infty$.

The Gromov product for the boundary points $\Psi^+(z_n), i(z) \in \partial_\infty G$ satisfies

$$\begin{aligned}(\Psi^+(z_n) \mid i(z))_P &\geq \liminf_{i,j\to\infty} (\tilde{\psi}_i(z_n) \mid z_j)_P \\ &\geq \liminf_{i,j\to\infty} \min\{(\tilde{\psi}_i(z_n) \mid z_n)_P, (z_n \mid z_j)_P\} - \delta\end{aligned}$$

where δ is the hyperbolicity constant of $\tilde{X}$. By Lemma 4.2, $(\tilde{\psi}_i(z_n) \mid z_n)_P \geq r_n - M$, where M depends only on K and δ. Thus, the above becomes

$$\begin{aligned}&\geq \liminf_{j\to\infty} \min\{r_n - M, (z_n \mid z_j)_P\} - \delta \\ &= \min\{r_n - M, a_n\} - \delta.\end{aligned}$$

Both r_n and a_n go to infinity with n, so the Gromov product $(\Psi^+(z_n) \mid i(z))_P$ goes to infinity with n, and $\Psi^+(z_n)$ converges to $i(z)$. □

Proposition 5.10 gives rise to a canonical map from the universal real tree to $\partial_\infty G$, so by [DKL26] we obtain the following corollary.

Corollary 5.11. *The set $\Psi^+(T_+) \subseteq \partial_\infty G$ only depends on a component of the BNS invariant.*

Proof of Theorem A. The identification of $\widehat{T}_\pm \setminus T_\pm$ with $E(T_\pm^{\mathrm{obs}})$ via Proposition 2.11 allows us to call a point of $E(T_\pm^{\mathrm{obs}})$ laminar if the corresponding point of $\widehat{T}_\pm \setminus T_\pm$ is laminar; we denote the set of laminar ends of $T_\pm$ by $E^{\mathrm{lam}}(T_\pm)$.

Let $\mathfrak{C}'_u$ denote the space $E(T_+^{\mathrm{obs}}) \setminus E^{\mathrm{lam}}(T_+^{\mathrm{obs}}) \sqcup E(T_-^{\mathrm{obs}}) \setminus E^{\mathrm{lam}}(T_-^{\mathrm{obs}})/\sim$, where an end $e_+ \in E(T_+^{\mathrm{obs}})$ is identified with $e_- \in E(T_-^{\mathrm{obs}})$ if: for any sequences $x_n \in T_+$, $y_n \in T_-$ with $x_n \to e_+$ and $y_n \to e_-$, we have that $\lim i(x_n) = \lim i(y_n) \in \partial_\infty G$.

By Proposition 2.11, $\mathfrak{C}_u$ and $\mathfrak{C}'_u$ are G-equivariantly homeomorphic. By Lemma 2.8, up to G-equivariant homeomorphism, $\mathfrak{C}'_u$ could have been defined using the intervals topology rather then the observers' topology. By Theorem 3.1, the intervals' topology of T_+, T_- only depends on the component Σ_0 of the symmetrized BNS invariant. Finally, Corollary 5.11 ensures that the laminar points are preserved within Σ_0 and the identifications made in defining $\mathfrak{C}'_u$ do not depend on the choice of u within Σ_0.

The action of G on $\mathfrak{C}_u$ preserves Bestvina–Feighn–Handel's laminations $L_{\mathrm{BFH}}(\varphi^\pm)$, and also $L(T_\pm) = \Lambda_{\mp\infty}$, so G preserves a pair of algebraic laminations. □

6. Relationship to Calegari–Loukidou's zippers

The methodology we used to construct $\mathfrak{C}$ is similar to—and inspired by—Calegari–Loukidou's *zippers* [CL26b]. If G is a hyperbolic group, a zipper is a pair of non-empty, path-connected, G-invariant subsets $Z^\pm \subseteq \partial_\infty G$ with $Z^+ \cap Z^- = \varnothing$. They were most interested in the dynamics of 3-manifolds, so their zippers were mostly subsets of S^2; the prototypical source of zippers $Z^\pm$ is the set of forward (resp. backward) endpoints of the lift to the universal cover of a pseudo-Anosov (more generally, quasi-geodesic) flow on a hyperbolic 3-manifold.

When $Z^\pm \subseteq S^2$, they proved that each of Z^+, Z^- is a real tree when topologized with an intervals-like topology. When $Z^\pm$ are real trees, say they are *minimal* if each of $Z^\pm$ is the convex hull of the G-orbit of any point. When $\partial_\infty G = S^2$, they showed every zipper contains a minimal zipper. In our setting, $\partial_\infty G$ is homeomorphic to the 1-dimensional *Menger curve* M^1 [KK00], and it is currently unknown whether arbitrary zippers on M^1 are real trees, though we suspect this is likely too good to be true.

Our construction of $\mathfrak{C}$ could have been given in the language of zippers: given a component Σ_0 of $\Sigma_s(G)$ for G hyperbolic free-by-cyclic with irreducible monodromy, let $(X^+, \psi^+), (X^-, \psi^-)$ be mapping tori associated to classes $u_+ \in \Sigma_0$, $u_- \in -\Sigma_0$, respectively. The universal covers $(\tilde{X}^\pm, \tilde{\psi}^\pm)$ are coarsely identified with G. Define $Z^\pm(\psi^\pm) \subseteq \partial_\infty G$ to be the set of endpoints of $\tilde{\psi}^\pm$. By Corollary 5.11, for any surjective $u\colon G \to \mathbb{Z}$ with $u \in \Sigma_0$, $Z^\pm$ is alternatively described as $\mathcal{R}_\pm(T_\pm(\varphi_u))$. The injectivity conclusion of Corollary 4.12 for $\mathcal{R}_\pm$ implies that $Z^\pm(\psi^\pm)$ are simply connected subsets of $\partial_\infty G$. As in [CL26b], we could consider each of $Z^\pm(\psi^\pm)$ topologized as a real tree and make our definition of $\mathfrak{C}$ from there. These zippers are minimal as the action of $\ker u$ on T_+ or T_- has dense orbits and no proper invariant subtree.

While Calegari–Loukidou focused on hyperbolic 3-manifolds, they gave constructions for hyperbolic groups from *uniform quasimorphisms* and *uniform actions*, see [CL26b]. We will describe their construction of zippers from the former and show that our semiflow zippers are minimal subzippers of their zippers—at least if certain constants are allowed to be sufficiently large.

Definition 6.1. Let G be a group, a function $u\colon G \to \mathbb{R}$ is a *quasimorphism* is there exists a number $D(u) > 0$, called the *defect*, such that for all $g, h \in G$,

$$|u(gh) - u(g) - u(h)| \leq D(u).$$

Some examples of quasimorphisms are homomorphisms and bounded functions; there are often many more, see [Cal09]. Calegari–Loukidou defined the notion of a *uniform* quasimorphism: a quasimorphism whose coarse level sets are coarsely connected, see [CL26b, Section 4] for details; see also [HK26]. A similar characterization of the BNS invariant $\Sigma(G)$ can be deduced from the Cayley graph definition of $\Sigma(G)$ [BR88, Corollary 6.3], see also [Str13]. Thus, we have the following fact: if $u\colon G \to \mathbb{Z}$ is a surjective homomorphism, it is uniform in the sense of Calegari–Loukidou if and only if $\ker u$ is finitely generated, i.e. $u \in \Sigma_s(G)$.

The following is the first definition needed to describe Calegari–Loukidou's zippers.

Definition 6.2. Let $u\colon G \to \mathbb{Z}$ be a uniform quasimorphism. A (C,k)-*staircase* for u is a sequence of group elements $a_n \in G$ such that

(1) $u(a_n) \leq u(a_{n+1}) \leq u(a_n) + 1$ for all n;
(2) $d(a_n, a_{n+1}) \leq C$ for all n;
(3) $u(a_{n+k}) > u(a_n)$ for all n.

We note that this definition differs slightly from their definition: in [CL26b], the C above always referred to a constant associated to the uniformity of the quasimorphism. Calegari–Loukidou prove that any (C,k)-staircase for u is a quasi-geodesic [CL26b, Lemma 4.7], so the *endpoint* of a staircase in $\partial_\infty G$ is well-defined when G is hyperbolic. For fixed $C, k \geq 1$, define $Z^+_{\mathrm{CL}}(u; C, k) \subseteq \partial_\infty G$ to be the set of endpoints of (C,k)-staircases for u. We will write $Z^-_{\mathrm{CL}}(u; C, k) = Z^+_{\mathrm{CL}}(-u; C, k)$.

Theorem 6.3 ([CL26b], Theorem 4.10)**.** *There are C, k large enough so that the sets $Z^\pm_{CL}(u; C, k) \subset \partial_\infty G$ form a zipper for G.*

Essentially, the first motivation for a zipper is as the forward (backward) endpoints of flow line of a pseudo-Anosov or quasi-geodesic flow. For a uniform quasimorphism, a staircase is a reasonable generalization of a flow line. We show that the zippers $Z^\pm$ arising from the universal real trees for $X^\pm, \psi^\pm$ are (minimal) subzippers of Calegari–Loukidou's zippers. We note that, with a few modifications, the proof also applies in the case of pseudo-Anosov suspension flows and is implicit in [CL26a; CZ26].

Lemma 6.4. *Let Y be a compact metrizable space, $f\colon Y \to Y$ a continuous function, and X the mapping torus of f with suspension semiflow ψ. Suppose further that $\pi_1(X)$ is Gromov-hyperbolic and $u\colon \pi_1(X) \to \mathbb{Z}$ the surjective homomorphism with $\pi_1(Y) \leq \ker u$. There exists $C > 0$ such that for each flow line $\gamma\colon [0,\infty) \to \tilde{X}$, there is a $(C,1)$-staircase for u with the same endpoint in $\partial_\infty \pi_1(X)$ as γ.*

Proof. The mapping torus structure defines a fibration map $\eta\colon X \to S^1$ with $Y = \eta^{-1}(0)$. Let $\tilde{p}\colon \tilde{X} \to X$ be the universal cover with lifted semiflow $\tilde{\psi}$ and $\tilde{\eta}\colon \tilde{X} \to \mathbb{R}$. Choose a fundamental domain X_0 of X by choosing a fundamental domain Y_0 of the $\ker u = \pi_1(Y)$ action on $\tilde{Y} = \tilde{\eta}^{-1}(0)$ and flowing it forward by one, that is, $X_0 = \{\psi_t(x) \mid x \in Y_0, 0 \leq t \leq 1\}$. We note that for any $g \in \pi_1(X)$ and any $x \in gX_0$, $u(g) \leq \tilde{\eta}(x) \leq u(g) + 1$ and if $x \notin p^{-1}(Y) = \tilde{\eta}^{-1}(\mathbb{Z})$, $u(g) = \lfloor \tilde{\eta}(x) \rfloor$.

Let $x \in \tilde{X}$ be any point not in $p^{-1}(Y)$ and consider the flow line $\gamma(t) = \tilde{\psi}_t(x)$. For each n, $\psi_n(x)$ is in some fundamental domain $a_n X_0$. This defines a sequence $a_n \in G$ that we claim is a staircase.

For any n, we have that

$$u(a_{n+1}) = \lfloor \tilde{\eta}(\psi_{n+1}(x)) \rfloor = \lfloor \tilde{\eta}(x) + 1 \rfloor = u(a_n) + 1.$$

Thus, a_n satisfies properties (1) and (3) of Definition 6.2 with $k = 1$.

For property (2), let $\tilde{x}_0 \in X_0 \subseteq \tilde{X}$ and fix some metric on X. Let D be the diameter of X_0 in the lifted metric on $\tilde{X}$. Fixing some word metric on G, the orbit map $G \to \tilde{X}$ defined by $g \mapsto g\tilde{x}_0$ is a K-quasi-isometry. Since $a_n X_0$ and $a_{n+1}X_0$ intersect, $d_{\tilde{X}}(a_n\tilde{x}_0, a_{n+1}\tilde{x}_0) \leq 2D$. Thus,

$$d_G(a_n, a_{n+1}) \leq K d_{\tilde{X}}(a_n\tilde{x}_0, a_{n+1}\tilde{x}_0) + K \leq K(2D) + K.$$

We have shown property (2) of Definition 6.2 with $C = K(2D+1)$. □

Proposition 6.5. *Let $u\colon G \to \mathbb{Z}$ be a group homomorphism in $\Sigma_s(G)$. For large enough C, k, the zipper $Z^+(\psi^+)$ we construct is a minimal subzipper of $Z^+_{CL}(u; C, k)$.*

Proof. The containment $Z^+(\psi^+) \subseteq Z^+_{\mathrm{CL}}(u; C, k)$ follows from Lemma 6.4. The action of G on the attracting tree T_+ has dense orbits and no proper invariant subtree, so $Z^+(\psi^+)$ is minimal. □

We now give an example of a (C, k)-staircase for a homomorphism $u\colon G \to \mathbb{Z}$ that is not asymptotic to a flow line of ψ^+. This will show that in general the containment $Z^+(\psi^+) \subseteq Z^+_{\mathrm{CL}}(u; C, k)$ is strict.

Proposition 6.6. *Let G be a free-by-cyclic group with $b_1(G) \geq 2$ and $u\colon G \to \mathbb{Z}$ a surjective homomorphism with finitely generated kernel (i.e. $u \in \Sigma_s(G)$). There exist C, k large enough so that the containment $Z^+(\psi^+) \subseteq Z^+_{CL}(u; C, k)$ is proper. In particular, for such C, k, $Z^+_{CL}(u; C, k)$ is not minimal.*

Proof. Let $u'\colon G \to \mathbb{Z}$ be distinct from u, but in the same component of $\Sigma_s(G)$, which exists because $b_1(G) \geq 2$ and $\Sigma_s(G)$ is open. There is some $g \in \ker(u + u')$ such that $u(g) > 0$. Then $u'(g) = (u + u')(g) - u(g) = -u(g) < 0$, thus the attracting fixed point of g^n in $\partial_\infty G$ cannot be in $Z^+(\psi^+)$ since $u'(g) < 0$. However, for large C, k, g^n is a subsequence of a (C, k)-staircase for u, and so the attracting fixed point of g is in $Z^+_{\mathrm{CL}}(u; C, k)$ □

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